

**IMAGE-BASED STUDENT IDENTIFICATION AND CLASSIFICATION
USING FACE RECOGNITION**

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Report

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Dedication

This work is a tribute to the unwavering support and encouragement we have received from our beloved parents. Their faith in us and continuous motivation have driven us to complete this project. We express our deepest gratitude to our dedicated teachers and supportive supervisor, whose time, effort, and invaluable moral support have guided us through this journey. They always believed in our potential. Our profound respect and appreciation are extended to our esteemed dean and research coordinator, whose persistent encouragement and insightful guidance have inspired us to excel in our research. Their wise counsel and protective guidance have been instrumental in shaping our work. We also thank the esteemed faculty members and lecturers of the department, whose expertise and willingness to assist us whenever needed have been immensely valuable. Their constructive feedback and encouragement played a significant role in refining our ideas. Above all, we acknowledge the blessings and grace of Almighty God, without whom we would not have achieved this milestone. May His abundant blessings be upon all who have contributed to this work and been part of our journey. With heartfelt gratitude and humility, we dedicate this work to our parents, teachers, supervisor, dean, research coordinator, faculty members, and all those who have supported us along the way. Your unwavering belief in us has made this accomplishment possible, and for that, we are forever grateful. May this work be a tribute to your unwavering support and contribute positively to the betterment of knowledge and understanding in our field.

ABSTRACT

In today's technology-driven world, the need for efficient and reliable student identification systems has become increasingly important, particularly in academic institutions where safety and accountability are critical. Traditional methods of identifying students, such as manual checking or ID cards, are often time-consuming, error-prone, and inadequate in handling large populations. To address these challenges, this research proposes an image-based student identification and classification system using face recognition technology.

The system leverages computer vision and machine learning techniques to capture, process, and analyze facial features for accurate recognition. By integrating CCTV cameras and Python-based tools such as OpenCV, Django, and NumPy, the system ensures real-time identification and classification of students. It also provides administrators with a reliable platform for distinguishing between lawful students and accused individuals during incidents of misconduct.

The proposed solution significantly reduces the manual effort required in student investigations, enhances security, and ensures efficient monitoring within the university environment. The results demonstrate that face recognition can be effectively applied in academic institutions to improve both operational efficiency and campus safety.

KEYWORDS

Face Recognition, Student Identification, Image Classification, Biometric Authentication, CCTV Surveillance, Artificial Intelligence, Computer Vision, Machine Learning, OpenCV, Python, Django, NumPy, Image Enhancement, Security Systems, Student Management, Accused

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CHAPTER 1:INTRODUCTION

1.0 Introduction

Face recognition is a fascinating and rapidly evolving field within the domain of computer vision and artificial intelligence. It involves the identification and verification of individuals based on their unique facial features. The ability to recognize faces has significant implications in various domains, including security, surveillance, human-computer interaction, and forensic investigations. With advancements in technology and the availability of vast amounts of visual data, face recognition has gained substantial attention from researchers and practitioners worldwide.

Therefore, this study investigates and identifies the development of a Face Recognition system and identifying which the accused was. Face Recognition is the process of identifying or verifying individuals based on their unique facial features. Facial recognition technology analyzes facial patterns, landmarks, or other distinguishing characteristics to match them against a database of known faces.

Image-based Student Identification and Classification using Face Recognition is also simplifying the current manual investigation taking process by automating the process using Face Recognition. This system also will allow the investigator to get accused in real-time.

1.1 Background OF The Study

The process of identifying a person from digitized facial photos is known as face recognition. Usually, the goal of an FRS is to produce a similarity score between two photos of faces. Finding important facial features, such the center of the eyes, for alignment and normalizing the face's appearance choosing a suitable feature representation, learning discriminative feature combinations, and developing accurate and scalable matching schemes. (Jain et al., 2012).

In the Face Recognition has always posed a serious challenge. The Face Recognition of the accused is an onerous task consuming a lot of time. An accurate automatic personal identification is

becoming very important nowadays for the operation of an increasingly electronically interconnected information society. Manual verification of the identity of an individual, through some captured Face Image, are not considered safe enough to make a decision or differentiate between accused students and a lawful.

CCTV cameras are thought to be one of the most crucial security component requirements since they help to solve this issue. These days, CCTVs are placed in every public area and are mostly utilized for crime prevention, safety, and other detection-based security measures. CCTV footage is the most important type of evidence used in courts. When law enforcement officials respond to a crime scene, they bring the video recording with them. When comparing the surveillance systems of various nations, we find that the United Kingdom has roughly 4.5 million cameras in use. Around 2010, there were 50,000 cameras installed in Sweden. The government of Poland installed merely 450 cameras in the city of Poznan, which resulted in a 60% reduction in drug cases and a 40% decrease in street clashes. China boasts the largest surveillance system in the world, with 170 million cameras dispersed around the country. By 2020, an additional 400 million cameras are scheduled to be linked, resulting in a threefold expansion of the system. Chinese authorities used its powerful military to locate and capture BBC reporter John Sudworth in under seven minutes. CCTV cameras network and facial recognition technology and put the accused behind the bar. (Bhatti et al., 2021).

Finding a specific individual from a variety of camera sources is made easier by facial recognition, which uses a number of subtasks, such as identifying a human in an image, extracting the face from it, processing the face to identify the unique features of the identifying the same individual by teaching the system to identify faces that are similar to yours. In the age of technology, locating and following someone is becoming an essential responsibility for various security purposes (Kumar et al., 2018).

Installation of CCTVs for constant monitoring of people and their interactions is a very common practice in most of the organizations and fields. For a developed country with a population of

millions, every person is captured by a camera many times a day. A large number of videos are produced and kept for a specific amount of time. Since constant monitoring of these surveillance videos by the authorities to judge if the events are suspicious or not is nearly an impossible task as it requires a workforce and their constant attention. We therefore have a requirement to highly accurately automate this process. Moreover, there is a need to show in which frame and which parts of it contain the unusual activity which aids the faster judgment of that unusual activity being abnormal or suspicious. (Singh et al., 2020).

An image classification technique for automobile air-conditioning leakage detection was proposed to use the mean weight least squares SVM technique to determine which part of the circuit is malfunctioning. Here, a type 2 adaptive system is used for enhancement and a Bayesian filter is used for noise reduction in the pre-processed image. In addition, a genetic algorithm was applied to the improved image in order to train the classifiers through feature selection. An expensive step in the image classification process is labeling the images. Although labeling takes more time and effort, it is vital for the categorization techniques training process.

1.2 Problem Statement

Although the number of students joining the university increasing year after a year and the violence of the student is becoming more and more it's difficult to seek the accused in the all classes and to check any students face to save that time and effort it needs to have one module that checks if the accused face exists in the students and which class attends and to best recognition is his/her ID.

In the current scenario, the process of checking a accused face and seeking into his/her class is time- consuming and inefficient. The process becomes harder to manage when the faculty contains more classes. Prevention of violence on the university is even harder to control. After each violence, if the face recognizes it difficult to recognize which class was what his/her ID was. organizations are not changing the way they investigate events. furthermore, the problems we have

traditionally based it are difficult on all sides. So to address the gaps mentioned above we implement a real-time monitoring system using Face Recognition on Image-based Student Identification and Classification using Face Recognition.

1.3 Motivation of the Study

The motivation of this study is to implement Image-based Student Identification and Classification using Face Recognition. Our university has challenges with identifying in terms of the accused, therefore, there is a need in developing this project, so we are going to implement this system which performs above work at the same time.

1.4 Objective of The Study

1.4.1 General Objective

The main objective of this project is to identify between an accused person (student) and a lawful before make a decision.

1.4.2 Specific Objectives

- To investigate between an accused person (student) and a lawful before make a decision.
- To design a system that facilitates seeking student on his/her captured Image face,
- To reduces the amount of time-consuming on checking students captured Image Face,
- To simplify the current manual searching taking procedure by automating the process using Face Recognition.

1.5 Research Question

- How do differentiate between an accused person (student) and a lawful before make a decision?
- How to design a system that facilitates investigation and recognize captured Image face?
- How to reduces the amount of time-consuming on checking students captured Image Face and minimize effort usage by getting that accused student?
- How to simplify the current manual searching taking procedure by using Face Recognition?

1.6 Scope of the Study

The scope of this project is taken to concentrate on Somali Organizations especially Jamhuriya University of Science and Technology (JUST), to control its operations related to the investigation, this study will carry out in Mogadishu- Somalia. As everything has a time and every time has its importance, this study will conduct between “December 2023” And “August 2024”.

1.7 Significance of the Study

The main importance of this project is to solve the traditional based way to a more accessible module to save time, effort, and the safety of the equipment of the university. so that when the project is done the following benefit will be realized: - every image takes by camera 4mp and more than that and distance about 5M or less will recognize that student quickly without taking more time. Helps the investigator to recognize an accused and a lawful student before take step of punishment.

1.8 Organization of the Study

- **Chapter One: Introduction-** This chapter introduces the study by presenting the research plan. It contains the background of the study followed by the statement of the problem, motivation of the study, the research objectives, the research questions, the scope of the study, the significance of the study, and the organization of the study as already discussed.
- **Chapter Two: Literature Review-** this chapter focuses on the previous kinds of literature about Face Recognition, Classification and any other relevant information.
- **Chapter Three- Research Methodology-** this chapter clearly points out the methodology used in this research which is to develop a software solution. it explains the hardware and software requirements to develop the system, and possible solution for the research question, methods, and concepts chosen.
- **Chapter Four: Design and Analysis-** in this chapter the researchers are going to demonstrate the logical design of the system that is to be developed by drawing the Entity

Relationship Diagrams (ERDs) and the Data Flow Diagrams (DFDs) to show the flow of the data within the system.

- **Chapter Five: Implementation and Coding-** This chapter confers the design and development of a system by using software tools and hardware devices. it also displays the most important code which makes a fundamental impact on the system functionality and screenshot of the system interface.
- **Chapter Six: Conclusion and Recommendation-** This chapter draws the conclusion of the research by suggesting the results of the research and finally provides the necessary future work recommendations related to the topic for future researchers in this area.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This chapter presents numerous kinds of literature on Image-based student Identification and classification using face Recognition. This literature was collected from different books, papers, articles, and studies related to the issue at hand. Specifically, the literature will concentrate on the identification methodologies, classification.

2.1 Biometric Identification

Measuring and examining each person's distinct physical or behavioral traits is the aim of biometric systems. The use of physical components with distinguishing qualities is the primary characteristic of biometric systems. Both physiological (fingerprint, iris, palm print, face, etc.) and behavioral (signature, walking, speech patterns, facial dynamics, etc.) biometric systems have been described in the literature. According to Taskiran et al. (2020), facial biometrics has emerged as a highly favored biometric data collection method due to its ability to capture biometric data without requiring the user's agreement or invading their personal space.

Biometric systems are required for person identification and verification systems due to the growing importance of human elements in next generation technology. Certain biometric systems rely on physiological properties that are static, like:

- Face Recognition
- iris
- fingerprint
- hand geometry
- voice Recognition

as well as systems that make use of behavioral traits, sometimes referred to as soft biometrics, like the signature, gait, walking rhythm, speech pattern, and facial dynamics (Taskiran et al., 2020).

2.2 Face Recognition

One of the primary biometric characteristics has been the face, which finds use in a wide range of fields such as education, marketing, finance, entertainment, security and law enforcement, and human-computer interaction. The primary application domains and their respective subdomains are enumerated. The identity, age, gender, race, and facial expressions that convey emotions and mental states are all revealed by a person's face. Engineering, psychology, and neurology are all involved in the multidisciplinary field of studying the human face and facial behavior. Face recognition differs from a number of other biometric features in that is especially appropriate for surveillance applications since it can be carried out in an unobtrusive manner and does not always require the subject's cooperation. Furthermore, face recognition is appropriate for behavioral biometrics since it can be based on both the face's dynamic and physical (static) properties. Face identification in unconstrained contexts is a difficult challenge because of differences in head attitude, illumination, age, and facial expression. Additionally, changes in look could result from facial hair, makeup, or accessories like scarves or glasses. Similarity between people (such as twins or cousins) makes facial recognition much more challenging (Taskiran et al., 2020).

2.2.1 Overview of face recognition

One of the most significant uses of biometrics-based authentication systems in recent decades is face recognition. Face recognition is a type of task pattern that involves classifying a face as known or unknown based on a comparison with photos of known individuals that are kept in a database. Face recognition is difficult because of the specific variability in information because of random variation across different people, including systematic variations from various factors such as lightening conditions and pose (Barnouti et al., 2016).

A facial recognition technique is an application of a computer for automatically identifying or verifying either an individual from a digital picture or a frame from a video source. The most organic method of biometric identification is this one. Facial recognition technologies have recently

developed into two areas and they are Facial metrics and Eigen's faces. Facial metric technology relies on the manufacture of specific facial features (the system usually looks for the positioning of eyes, nose, and mouth and the distances between these features). The region of the face is resized to a predetermined, set size (e.g., 150–100 points). The term "canonical image" refers to this image of a standardized face. Following that, a face template contains the computed and stored facial metrics. The typical size of such a template is between 3 and 5 KB, but there exist systems with the size of the template as small as 96 bytes (Bhattacharyya et al., 2009).

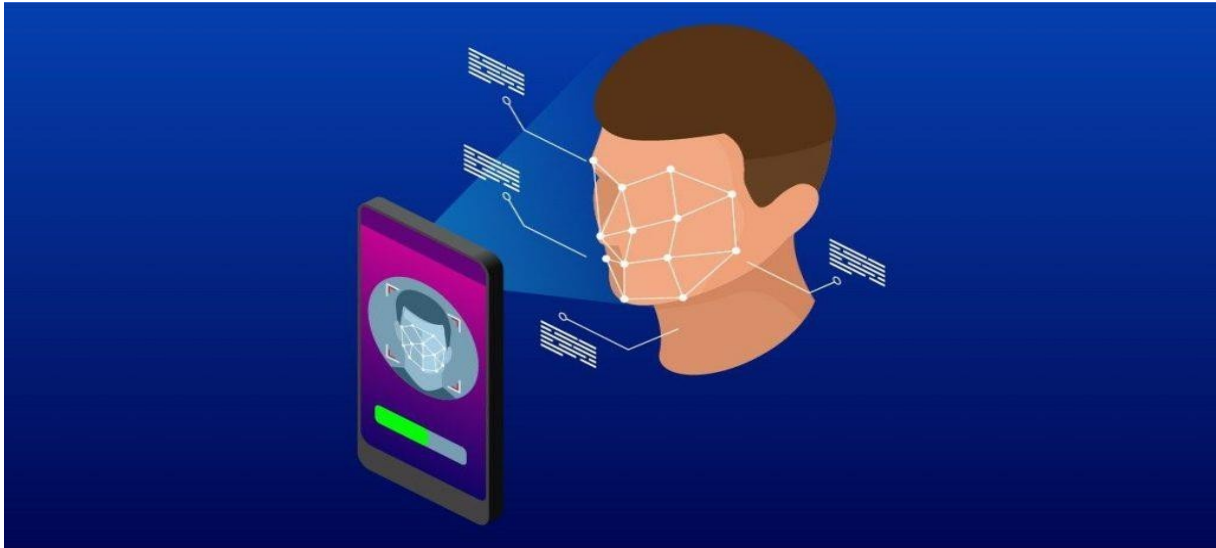


Figure 1:2:1: Face Recognition

2.2.2 Taxonomy of face recognition

The two primary categories of face recognition systems seen in the literature are image-based and video-based techniques. Image-based systems use a person's physical characteristics to attempt to identify them. However, physical appearance as well as variations in appearance over time or facial dynamics are used by video-based systems. The general taxonomy of the literature on face recognition (Taskiran et al., 2020).

Three primary categories can be used to group image-based facial recognition (FR) techniques: i) appearance-based (or holistic) techniques, ii) model-based techniques, and iii) texture (local appearance) based techniques.

FACE DETECTION - The computer vision system tool box present in the MATLAB is very helpful for detecting faces from the CCTV footage. The computer vision tool box provides the object detection process and gives a custom detector. The people's face can be easily recognized by using the people detector. Some of the histogram based classifiers are following Histogram of oriented gradients (HOG) features and a trained support vector machine (SVM) classifier. The output of the face detection method is always a cropped face image (Celine & others, 2019).

To classify the faces, a graph (i.e., kernel) It can also be used. It is the natural model of the relationship between faces. The nodes in this model stand in for faces. The potential interactions between nodes are represented by the edges. Then, machine learning methods such as support vector machine, kernel ridge regression, artificial neural networks or the graph-based semi-supervised learning methods can be applied to this graph to classify or to recognize the faces. Supervised learning techniques include kernel ridge regression, support vector machines, and artificial neural networks. The graph-based semi-supervised learning methods are the semi-supervised learning methods. Please note that the kernel ridge regression method is the simplest form of the support vector machine method (Tran et al., 2020).

There are two primary categories of video-based face recognition techniques: set-based techniques and sequence-based techniques. Set based methods treat the frames of a video sequence as a collection of images, without paying attention to the temporal order of the frames. Sequence-based approaches, on the other hand, exploit the frames while maintaining their temporal order. Hence, the dynamics of the face over time also plays a role in the recognition of the person It is very difficult to give a clear-cut taxonomy of all the work on face recognition in the literature. Hence, the proposed taxonomy in Fig. 1 is a coarse grouping of the methods in the literature and the algorithms in some groups may have overlapping properties (Taskiran et al., 2020).

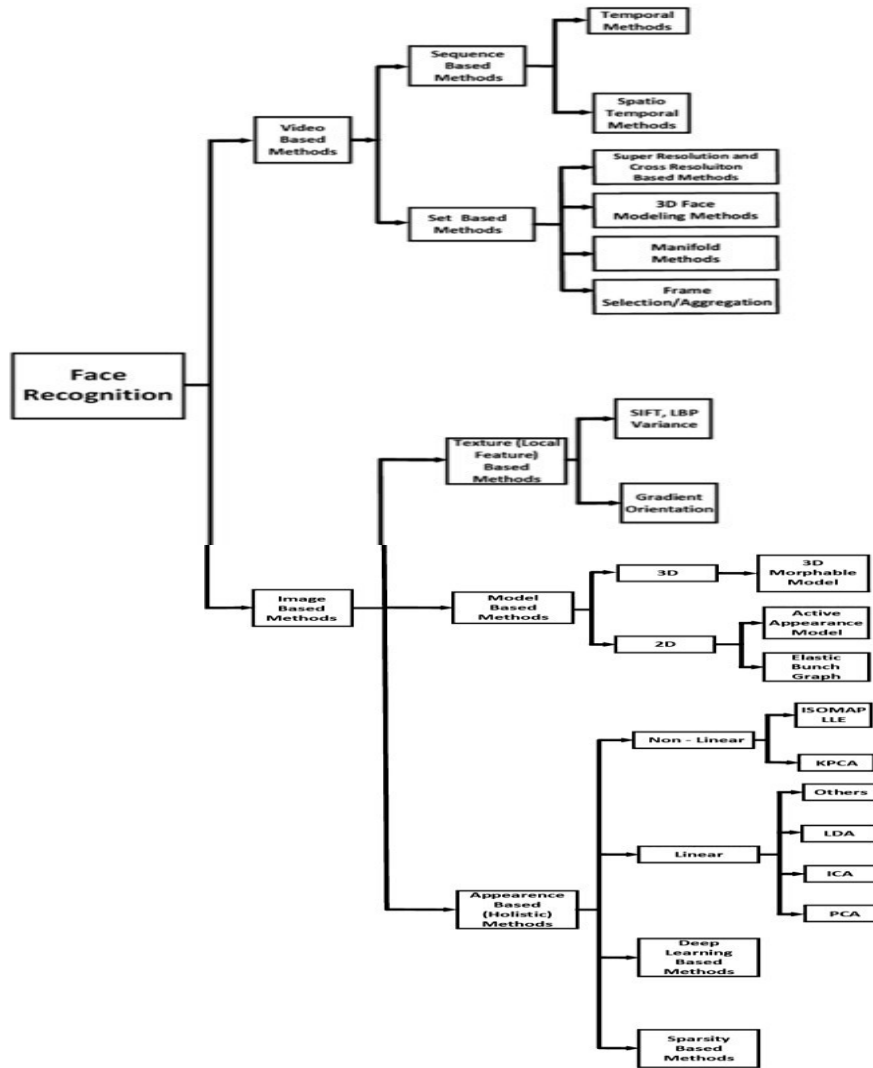


Figure 2:2:2: face recognition

2.2.3 Main steps of face recognition

Face recognition can be viewed as either a verification or identification challenge. i) The face is captured in an input image or video.

Face anti-spoofing module ensures the security of the system by employing presentation or adversarial attack detection (via liveness tests etc.).

A face or facial landmark is identified in every frame of the video or image.

Pre-processing is the process of applying several techniques to an image or video, such as

alignment, noise reduction, contrast enhancement, or video frame selection.v) Extraction of facial

features from a picture or video. While video-based methods utilize set-based or sequence-based feature extraction techniques, image-based methods use holistic, model-based, or texture-based techniques.

Face identification or verification is performed.

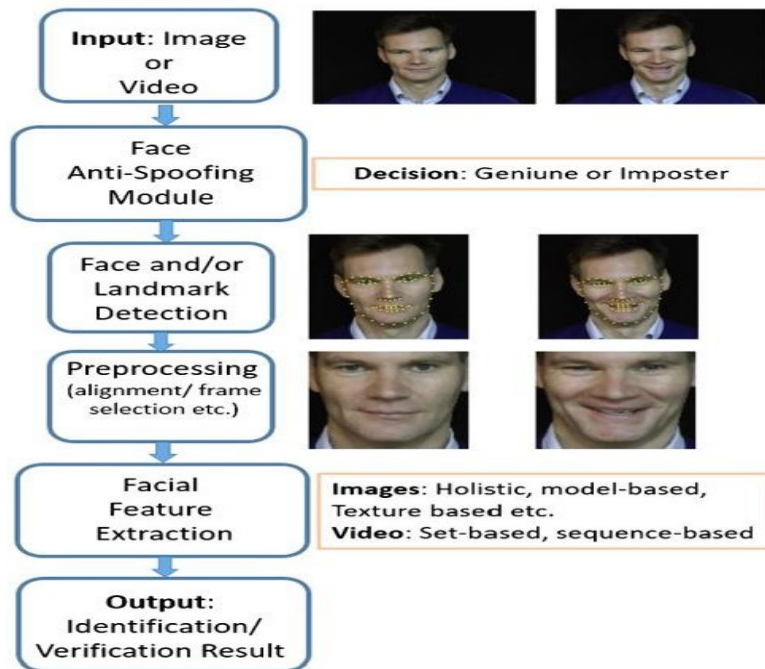


Figure 3:2:3: steps of Face Recognition

2.3 Iris recognition

Face This recognition method uses the iris of the eye which is a colored area that surrounds the pupil. Iris patterns are unique and are obtained through a video-based image acquisition system. Every iris structure has an intricate pattern. A combination of distinct features called the corona, crypts, filaments, freckles, pits, furrows, striations, and rings may be present. the iris pattern is taken by a special gray scale camera at a distance of 10- 40 cm of camera. Once the gray scale image of the eye is obtained then the software tries to locate the iris within the image. The software covers the iris with a net of curves if an iris is detected. The software generates the iris code based on the point-to- line darkness. Two influences need to be considered in this situation.

First, the overall darkness of the image is influenced by the lighting condition so the darkness threshold used to decide whether a given point is dark or bright cannot be static, it must be



dynamically computed according to the overall picture darkness. Secondly, the size of the iris changes as the size

Figure 4:2:4: Iris

2.4 Fingerprints

Fingerprints are widely used in the personal identification system. In the era of information technology, fingerprint identification is popular and widely used worldwide, not only for anti-criminal but also as a key technique to deal with personal affairs and information security (Yang & Nanni, 2011). Any or all of the finger's friction ridges can leave an impression on a fingerprint. A elevated area of the palmar (palm), digit (fingers and toes), or plantar (sole) skin made up of one or more interconnected ridge units of friction ridge skin is called a friction ridge. Occasionally, these ridges are referred to as "dermal ridges" or "dermal." The traditional method uses ink to get the fingerprint onto a piece of paper. After that, a conventional scanner is used to scan this piece of paper. Now in the modern approach, live fingerprint readers are used. These are founded on principles related to optics, heat, silicon, or ultrasonics. It is the oldest of all biometric techniques. An optical fingerprint reader is the most common at present. They are based on reflection changes at the spots where finger papilar lines touch the reader's surface. All the optical fingerprint readers comprise the source of light, the light sensor, and a special reflection surface that changes the

reflection according to the pressure (Bhattacharyya et al., 2009).



Figure 5:2:5: Fingerprint

2.5 Hand geometry

Hand geometry has emerged as one of the most widely used biometric access control technologies in recent years, accounting for about half of the physical access control industry. Some features related to a human hand, e.g., length of fingers, are relatively invariant and peculiar (although, not unique) to each individual. The subject must cooperate with the image acquisition equipment in order to take frontal and side view pictures of their palms spread outward on a panel. The registration of the palm is accomplished by requiring the subject's fingers to be aligned with a system of pegs on the panel which is not convenient for subjects with limited flexibility of palm (Jain et al., 2006). It is based on the fact that nearly every person's hand is shaped differently and that the shape of a person's hand does not change after a certain age. These methods involve estimating the hand's surface area, thickness, width, and length. Various methods are used to measure the hands- Mechanical or optical principle There are two sub-categories of optical scanners. Devices in the first category produce a bitmap image of the hand in black and white. This is easily done using a source of light and a black and white camera. The bitmap image is processed by computer software (Bhattacharyya et al., 2009).



Figure 6:2:6 Hand recognition

2.6 Voice Recognition

Face Voice is also a physiological trait because every person has a different pitch, but voice recognition is mostly focused on the analysis of a person's speech pattern, which is categorized as behavioral. The vocal traits that generate speech are the main emphasis of speaker verification, not the speech's actual sound or pronunciation. The dimensions of the vocal tract, mouth, nasal cavities, and other speech processing mechanisms in the human body affect the vocal features. It doesn't require any special and expensive hardware (Bhattacharyya et al., 2009).

As reported by (Subha, 2017), Voice or speech recognition is the capacity of a machine or program to understand and execute spoken orders, or to take in and interpret dictation. Although each person has a unique pitch, voice recognition is primarily dependent on behavioral analysis of speech patterns, which makes voice recognition a physiological attribute as well.

2.7 Image Classification

An important area of study for artificial vision systems that has attracted a lot of attention recently is picture classification. The purpose of this field is to categorize an input image according to its visual content. At the moment, the majority of individuals used manually created features to give an image a specific description. The final conclusion was then reached by applying learnable classifiers, like random forests and decision trees, to the extracted features. The problem arises when large numbers of photos are concerned (Singh & Singh, 2020).

2.8 CCTV Camera

Body-worn video cameras (BWVCs) and closed-circuit television (CCTV) have proliferated in the police field. The significance of determining optimal procedures for these two technologies has increased due to their extensive implementation. The research community has been working to identify these best practices, and in the last few decades, bodies of knowledge have emerged for both CCTV and BWVCs over recent decades. Given its earlier emergence, research on CCTV is

more developed(Piza, 2021).

There has been extensive There is a dearth of studies on the investigative utility of closed-circuit television (CCTV), but plenty on its effectiveness in preventing crime. In order to determine how frequently and how conditions affect the usefulness of CCTV evidence, this study analyzed 251,195 offenses that the British Transport Police had reported as having occurred on the British railway network between 2011 and 2015. Investigators had access to CCTV in 45% of cases, and in 29% of those cases (or 65% of those when it was available), it was deemed to be beneficial. Effective CCTV was linked to noticeably higher odds of crimes being solved for every category of crime, with the exception of fraud and the presence of narcotics or weapons. Serious crimes were more likely to have available images, but incidents that happened at odd hours or in particular kinds of places were less likely to have images. CCTV seems to be an effective investigative tool for many different kinds of crime, despite the fact that this research was restricted to offenses on railroads. The amount of public spaces that CCTV cannot cover is one of the main problems limiting its usefulness. There are some suggestions made for improving the use of CCTV (Ashby, 2017).

2.9 Steganography

Steganography is the art and science of message transmission through obfuscation. This is accomplished by disguising the existence of the sent information, or burying it within other information. The Greek terms "stegos," which means "cover," and "grafia," which means "writing," are the source of the word "steganography," which means "covered writing." With image steganography, data is concealed within pictures. Information concealment has a lengthy history, both as a theory and a practice. Greek historian Herodotus describes a nobleman named Histaeus who had to get in touch with his son-in-law in Greece in his book Histories. One of his most reliable slaves had his head shaved by him, and the message was permanently inked on the

slave's scalp. The slave was sent away with the secret message once his hair had grown back (Kavitha et al., 2012).

Steganography is a subfield of cryptography that aims to conceal messages from others without giving the impression that a message is present. Any type of cover file can be used to apply stenographic techniques, albeit the most common types these days are picture, sound, and video file archives. Comparably, any type of information can be hidden: text, photos, videos, sounds, etc. Currently, there are two approaches being used to implement stenographic algorithms: those that operate in the spatial domain, changing the desired properties of the file directly, and those that operate in the transform domain, making multiple adjustments to the cover image before concealing data(Roque & Minguet, 2009).

2.10 Image enhancement

One of the most crucial aspects of digital image processing is image enhancement. It is among the trickiest and most significant tasks in the processing of digital images. Techniques for image enhancement are applied to raise an image's visual quality. Nowadays, the majority of applications, including radiography, MRI, nuclear medicine, ultrasound imaging, tomography, cardiograph, fundus imaging, and so on, use medical imaging. The two main issues with medical images are contrast and image quality. Image enhancement clarifies the image for computer analysis or human viewing. While image augmentation doesn't increase the data's inherent information content, it might highlight the features of interest to detect the objects in a simple and efficient manner (Vasuki et al., 2017).

Image enhancement is a powerful tool which is widely used in computer vision. It is a popular sub-specialization of image processing. The practice of enhancing an image to make it visually more appealing and comprehensible for upcoming automated image processing operations including analysis, detection, segmentation, and recognition is known as image enhancement. Improving interpretability or perception of data in images can be thought as main goal of image enhancement

process so that human viewers or future automated image processing can benefit from the new enhanced image. When dealing with image enhancement it must be kept in mind that image enhancement implementation doesn't have effect upon the intrinsic information content of the data. What is effect being to increase the dynamic range of the intensity values of image. Over the years a variety of image enhancement methods have been proposed with the developments in image processing field. These methods have been independently studied for several different applications, resulting in a large body of research. Although there is a wide range of methodologies, classification and comparison of techniques become difficult because each technique is often designed for specific applications and not necessarily for specific types of problems or data. However, the approaches differ in information on which enhancement relies. Briefly stated, the researchers or scientists who intend to enhance images should establish the connection between the changes in the images/targets of enhancement and the type of enhancement methods which can most appropriately be applied. (Irmak & Ertas, 2016).

Enhancing an image simply means making it easier for human viewers to interpret or perceive the information included in it, as well as better supplying input for various automated image processing methods. Many methods are available for improving a digital image without compromising its quality. The two categories into which the improvement techniques can be generally separated are as follows: 1. Spatial Domain Methods 2. Methods in the Frequency Domain When using spatial domain approaches, we work directly with the pixels in the image. The desired improvement is obtained by manipulating the pixel values. The picture is initially moved into the frequency domain when using frequency domain techniques. This indicates that the image's Fourier Transform is calculated first. After completing all enhancement procedures on the image's Fourier transform, the resulting image is obtained by performing an inverse Fourier transform. Image enhancement is applied in every field where images are ought to be understood and analyzed. For example, medical image analysis, analysis of images from satellites etc. In this

section we briefly describe the various image enhancement techniques (Vishwakarma & Mishra, 2012).

CHAPTER 3: RESEARCH METHODOLOGY

3.0 Introduction

This chapter is related to the presentation of the study and it contains the research methodology chosen to achieve this research study. It also consists of a system description, system architecture, system features, and system requirement specification.

3.1 System Description

Image-based Student Identification and Classification using Face Recognition are resourceful in giving an effective and reliable verification of student's accused over the existing image captured. it involves to know if image captured is student in the or Someone who is not and to avoid to mixed accused person (student) and a lawful.

The investigator will scan the the image captured by the camera to know his/her class, his/her ID and so and also facilitates to take decision quickest to know which person is accused or which one is lawful and also allow to reduce to happen wrong actions.

3.2 System Features

To get accused person's: This helps the investigator if the camera captured full face to know if this face is part of Student and also know which class and what is his/her ID

Is.To get Respected Person's: This help investigators an image captured partial of the face and not full face we will try to know this part of face which students is to be a passible to become it get more students that is look after for this image.

3.3 System Requirement Specification

3.3.1 Software Requirement

Python: python is programming language to set up or program this project, Python is an interpreter, object-oriented, high-level programming language with dynamic semantics. Its dynamic typing and dynamic binding, along with its high-level built-in data structures, make it an appealing language

for rapid application development and for usage as a scripting or glue language to join pre-existing components.

Because of its straightforward, simple syntax, Python prioritizes readability, which lowers the cost of program maintenance. Python's support for packages and modules promotes code reuse and program modularity. The Python interpreter and the vast standard library can be downloaded and used for free on all major platforms, either in source or binary format. Furthermore, Python is a dynamic programming language that works with a variety of paradigms:

I. Procedural programming

II. Object-oriented programming

Django: was introduced in 2003 when Adrian Holovaty and Simon Willison began using Python to build applications. The motivation for developing Django was the need to get applications done on a schedule of days or even hours. Two years after starting to create a web application framework, in 2005, it was released as open source. Today, 10 Django has several tens of thousands of users and development (Ghimire, 2020).

Django official Logo Because Django is a web application framework based on Python, you need Python to use it. Supported versions of Python are 2.6.5 and 2.7, but Django also offers experimental support from version 3.2.3 to version 3.3. Although Python includes a lightweight testing server, an Apache server and the mod_wsgi package are required for the future Django-based website to function properly. Django officially supports PostgreSQL, MySQL, Oracle and SQLite databases (Ghimire, 2020).

Django also loosely applies MVC architecture. It becomes evident in Django, parts of the applications you created. In essence, models.py, views.py and urls.py, and the HTML template. Models.py contains a description of the information and provides functions for creating, retrieving, updating and deleting the data. View.py transmits the site content HTML template. The

role of the `urls.py` file, in turn, is to configure which view file is called. With this structure and file functions, editing an application does not require editing many files, which in turn saves time (Ghimire, 2020).

NumPy: is Multidimensional arrays, or tensors, can be easily stored and accessed with a data format that facilitates a broad range of scientific computations. It is composed of a pointer to memory plus metadata, such as "data type," "shape," and "strides," that are used to analyze the data stored there (Harris et al., 2020).

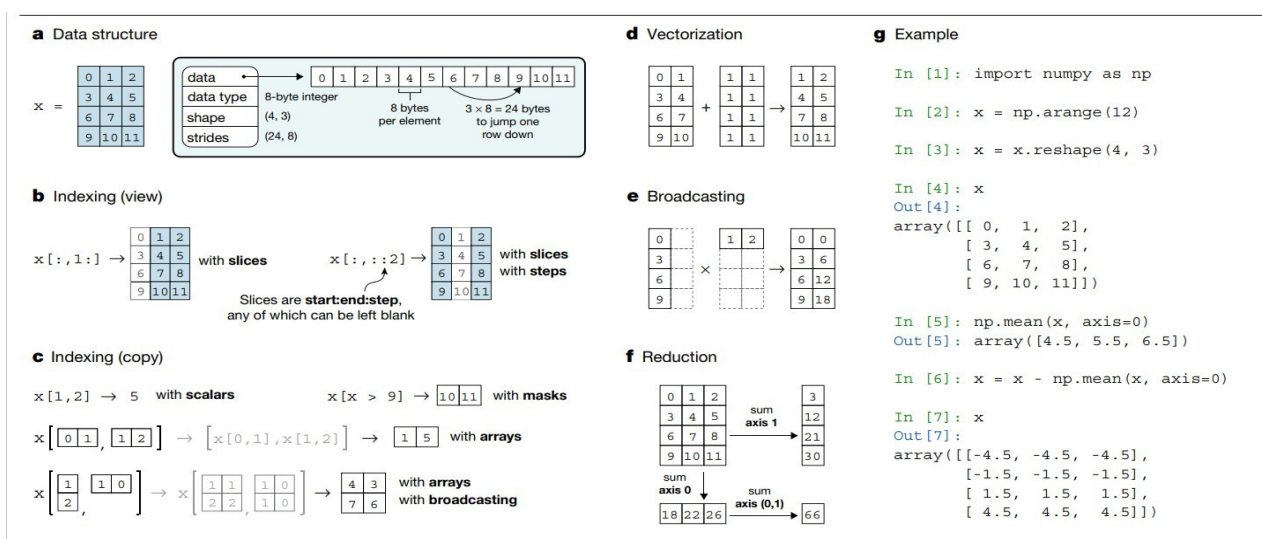


Figure 7:3:1: Numpy

OpenCV: is an open-source vision and machine learning software library written in Python. There is an extensive collection of reliable algorithms in the library. We'll take into consideration a few techniques for this case, such as face detection, face recognition, and similarity searches using a picture database. It can operate on a variety of operating systems and supports a wide range of programming languages. The other libraries will be employed in the proposed software system to gate a suitable development (Ismael & Irina, 2020)

steganography lsb: is the process of enclosing data inside an image and then removing it to view it on the other side. However, using images carries the risk that data may be concealed inside of them, as they are more likely to be deciphered by unauthorized individuals who may be in

possession of sensitive information. Emojis, also known as emoticons, are social conventions for expressing human behavior, emotions, and other states; unlike photos, they help keep human expressions hidden from prying eyes. Emojis can be handled more quickly and efficiently than photographs because of their small size and resolution, which also results in them consuming less memory space (Pabbi et al., 2021).

Because of its large payload capacity, the LSB technique is employed in this research to conceal the data inside an emoji. However, if the Emojis are owned by people or groups, information can be gleaned from them. Before hiding the data inside the emoji, using encryption is suggested as a solution to the problem. Since AES has shown to be more effective, safe, and efficient than DES and RSA, it is utilized as the encryption standard instead of these two (Pabbi et al., 2021).

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.0 Introduction

To build an innovative system that automates identifies student faces, and which done wrong thing and identifying which class goes. Our system uses University CCTV camera's captured images. So, this chapter focuses on the analysis and design of a developing Image-based Student Identification and Classification using Face Recognition, which is an important stage when developing systems such as Web Application.

4.1 Existing Approaches

Traditionally, In the university system, not all gives ability to differ from accused student's and lawful student's. The student simply done what they want without worrying to identify so quickly in the university building.

The traditional way has a number of problems which include the following:

- Traditional way is difficult to accurately differentiate between an ac accused student and an lawful student before taking decision.
- Traditional way requires a lot of time to recognize the face and which class he/she attendance.

4.2 The Purposed System

This system is designed to monitor students' face recognition and solve the drawbacks of old way. It has two functions including face recognition and suspect face. First, the system is recognizes face fully without suspecting others when the camera captured Image for fully faces and the camera took photo distance about 5 meter and about quality of 8MP or greater that Image can recognize 90%. The suspect faces when camera took part of the face like one side such as one eye, one ear, part of nose and part of chin when have this side we will list number of students suspected to done by this act.

4.3 System Requirements

A requirement is a written document of a characteristic that a new system must have. In this section, researchers discuss requirements for the Image-based Student Identification and Classification using Face Recognition and classify the requirement into two parts Functional Requirements and Non- Functional Requirements.

4.3.1 Functional Requirements

A functional requirement is a requirement that describes an activity of your system. The following are the requirements that the proposed system should functionally achieve.

Admin registration: the system is capable of registering an authenticated user to allow access to the system.

Training System: The system will allow you to train students face Image to recognize suspected face.

To Identification: is to pass image captured by the camera and return student ID, Name, Class and so.

4.3.2 Non-Functional Requirements

A non-functional requirement is one that specifies how to determine whether a functional requirement has been met. The following are the non-functional requirements that the proposed system must meet:

- Security: The system should be secure in order to ensure the privacy of its users.
- Availability: The system is accessible via the internet and can be accessed at any time and from any location.
- Privacy: The system can only be used by administrators and authorized users.
- User Friendly: The system is simple and interesting.

4.4 Feasibility of Study

4.4.1 Operational feasibility

The proposed system is divided into three main parts; the first part consists of training data. The second segment consists of a the second is testing therefore the third segment is a website that manages users to allow access.

4.4.2 Technical feasibility

The technical resources needed to develop a face recognition System using CCTV camera researchers needed to install Visual Studio Code, and computers running Linux operating systems OS also install django Server for mysqlite3 database, Firefox Browser or Brave Browser for security and Chrome Browser (Optional), all these technical resources needed are detailed explanation in chapter 3.

4.5 System Design

System design is the process of designing the elements of a system such as the architecture, modules, and components, the various interfaces of these components, and therefore the data that goes through that system. The objective of the system is to process of creating or alter new systems, along with the processes, practices, models, and methodologies used to develop them. The system design describes activities that involve specifying how the proposed system operates and works. We use UML to create system design and its various components. UML diagrams are used and analyzed using diagrams and activity diagrams Image-based Student Identification and Classification using Face Recognition

- Create User Account

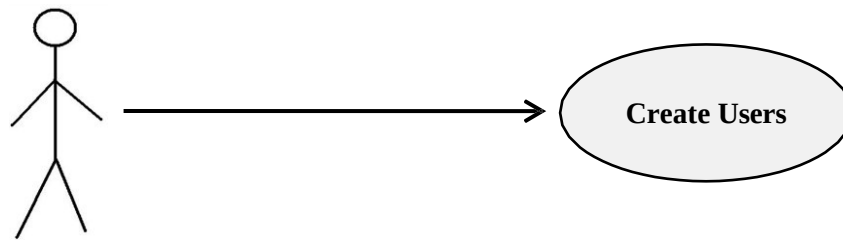


Figure 9:4:2: Create User Account

Brief description: This function allows the system administrator to create a new user.

Basic flow

- 1) The admin accesses the system and chooses the new account function to create a user account.
- 2) The system displays a registration form, including a Username, Password, User type and email.
- 3) The Administrator fills out the registration form and submits it.

- Training Model

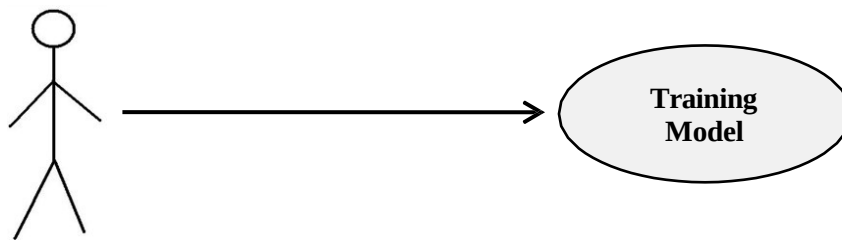


Figure 10:4:3: Use Case Diagram for Investigator

Brief description: This function allows the system administrator to train all student face to the system. Basic flow

- 1) The admin accesses the system and chooses the Train form.
- 2) The system displays a training model form and choose the student's face then it submits to the system.

- Testing Model

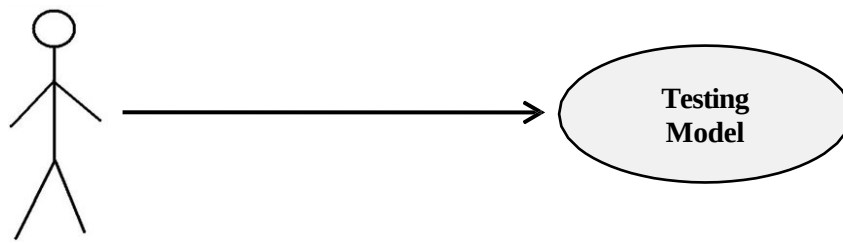


Figure 11:4:4: administrator testing modal

Brief description: This function allows the system administrator to test suspected student face to the system.

Basic flow

- 1) The admin accesses the system and chooses the Test form.
- 2) The system displays a test model form and choose the student's face then it gives the result of suspected face

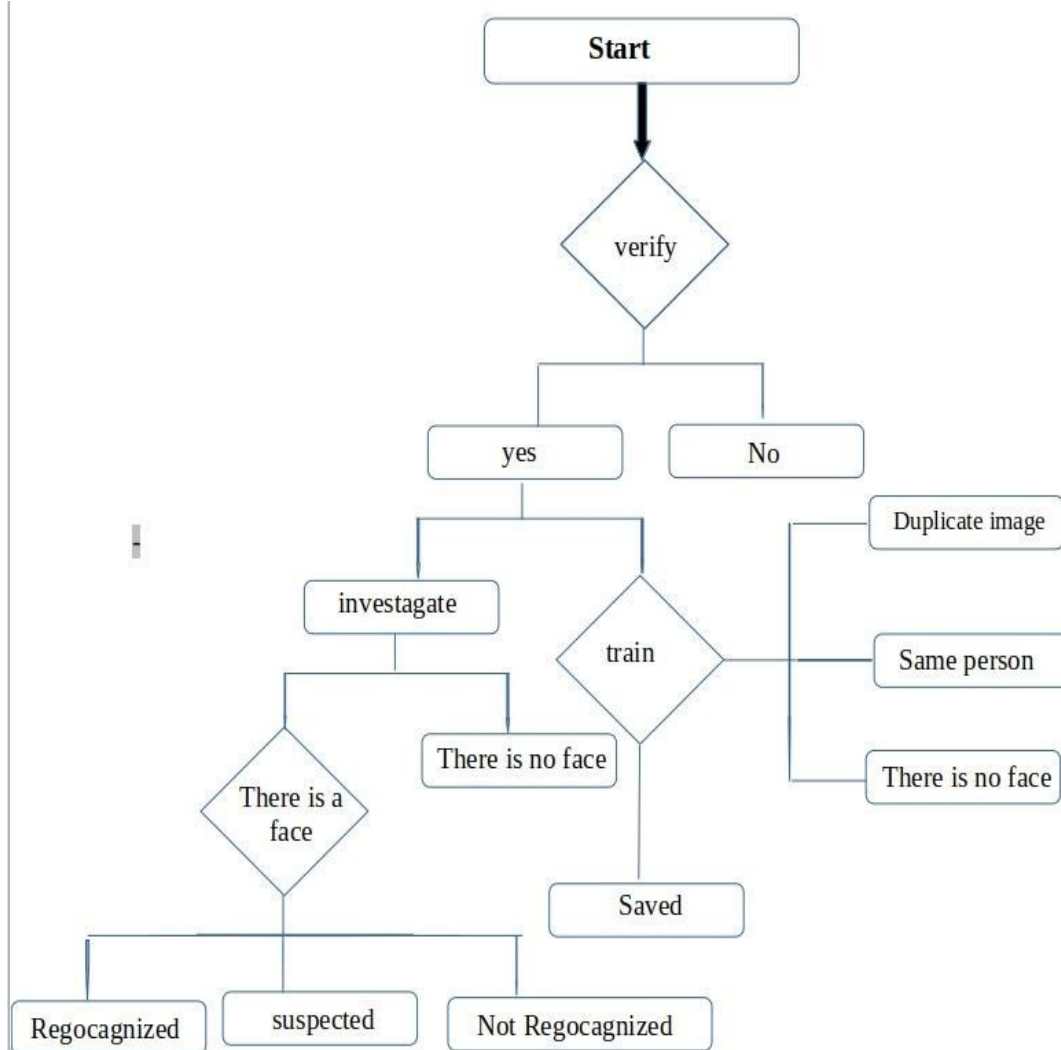


Figure 13:4:6: 1 Investigator testing modal

4.6 Database Design

Database design is the process of organizing data in a ascertain what information needs to be saved and how different data items connect to one another is known as database design. They can start fitting the data into the database model using this information. In order to create a database that satisfies user needs, appropriate database architecture is essential. Creating logical and physical representations of the intended database system's designs is the primary goal of database design.



examauthdb usertbl	
userid	int(11)
username	varchar(50)
userpass	varchar(50)
usertype	varchar(50)

5.0

6.0

7.0

8.0

9.0

10.0

11.0

Figure 14:4:7: User table database design

CHAPTER 5: IMPLEMENTATION AND CODING

5.0 Implementation and Testing Environment

The implementation and testing of the image-based student identification and classification system using face recognition were conducted in a carefully controlled environment to ensure accurate results and reliable performance. The development environment comprised a combination of hardware and software components chosen to support the functionalities required for the system.

5.1 Snapshots Of the system

Below are snapshots showcasing the key functionalities and user interface of the implemented image- based student identification and classification system using face recognition?

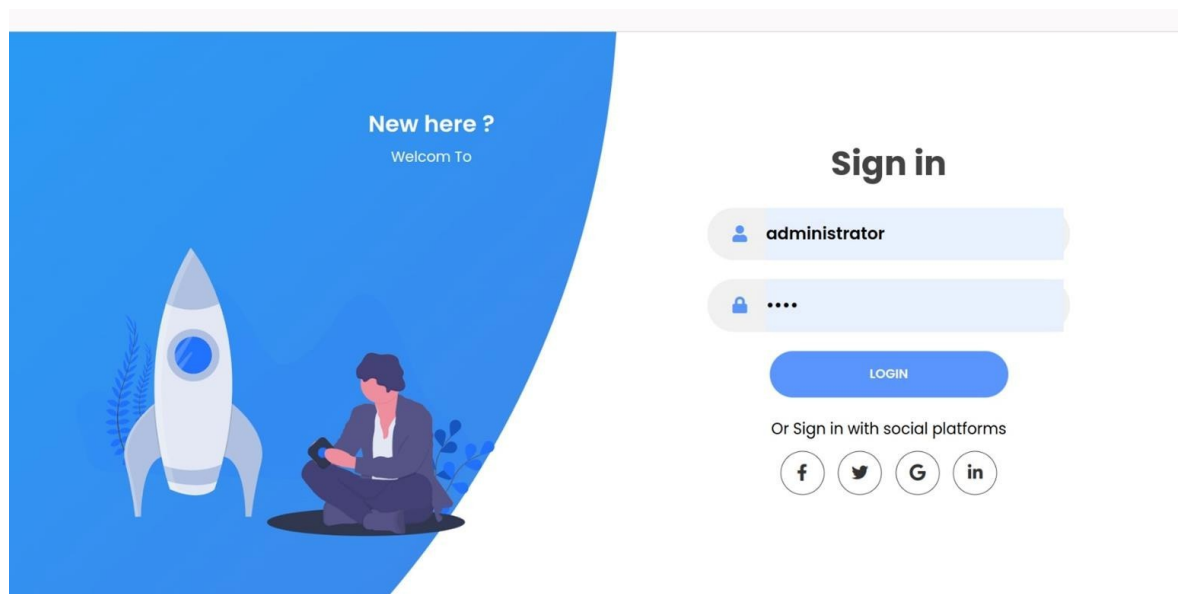


Figure 15:5:1 Sign up page

- Users access the system through a secure login page where they input their credentials to gain access to the system's functionalities.

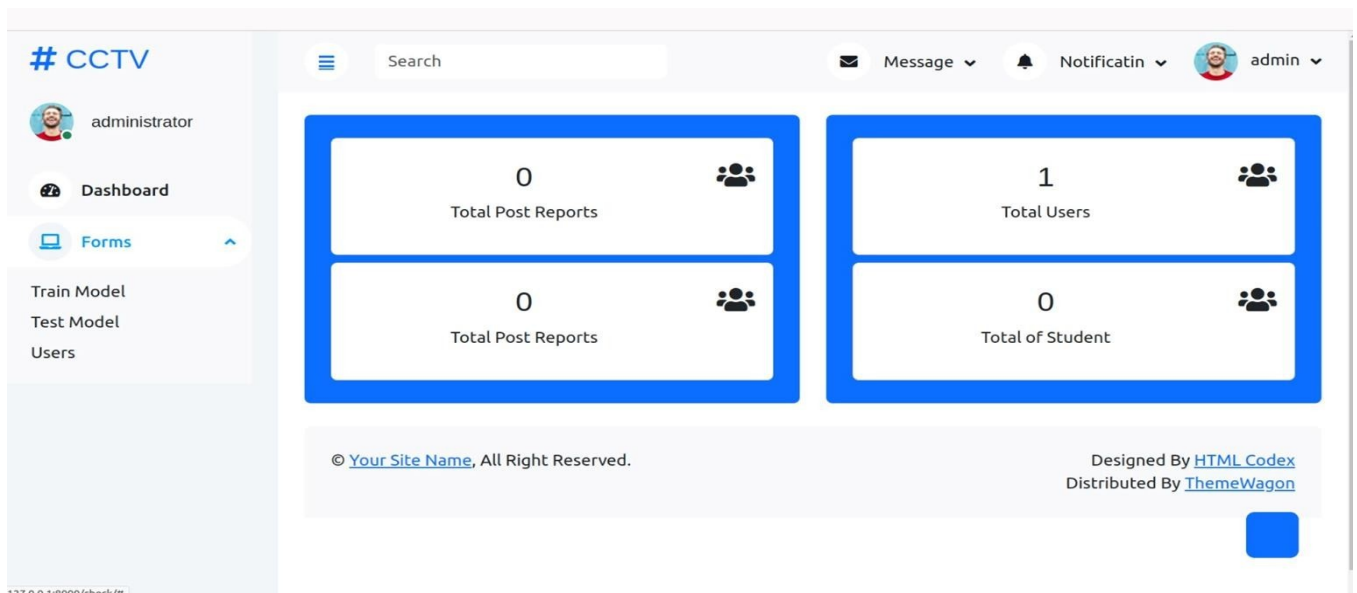


Figure 16:5:2: Dashboard Page

- Upon successful login, users are presented with a dashboard displaying essential statistics, such as the total number of registered students, the number of accused individuals, and the number of lawful students.

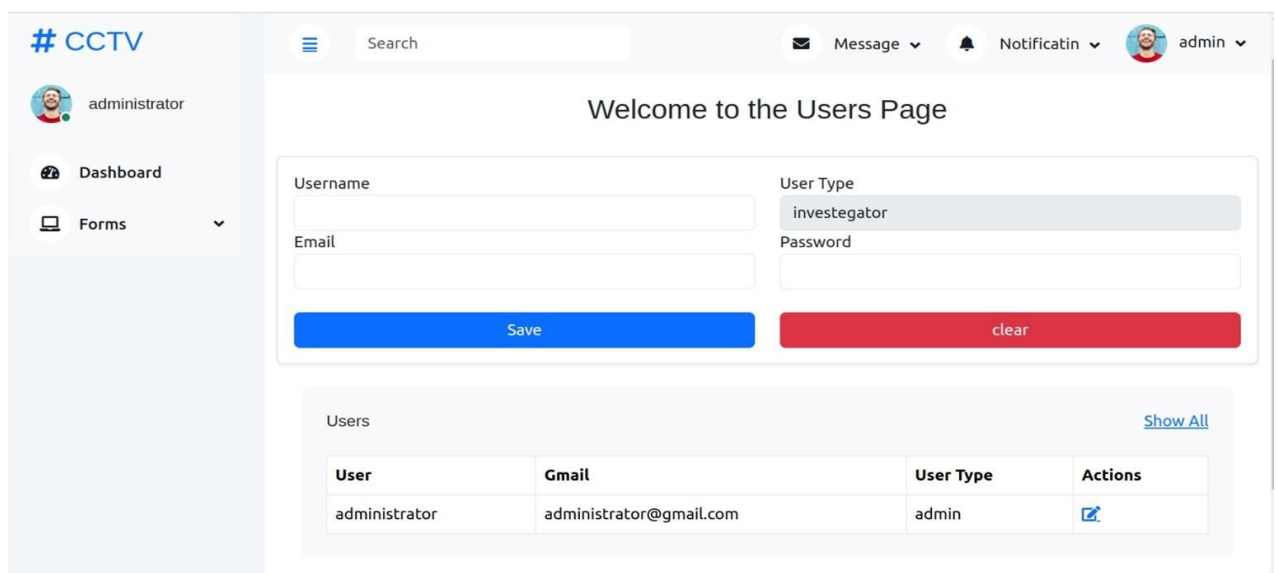


Figure 17:5:3: Users page

- Administrators can register new users through a dedicated registration page. This snapshot showcases the user registration process facilitated by the admin, ensuring controlled access to the system.

- This snapshot showcases the training page where users can upload images to train the system. The system expects image uploads to enhance its facial recognition capabilities.

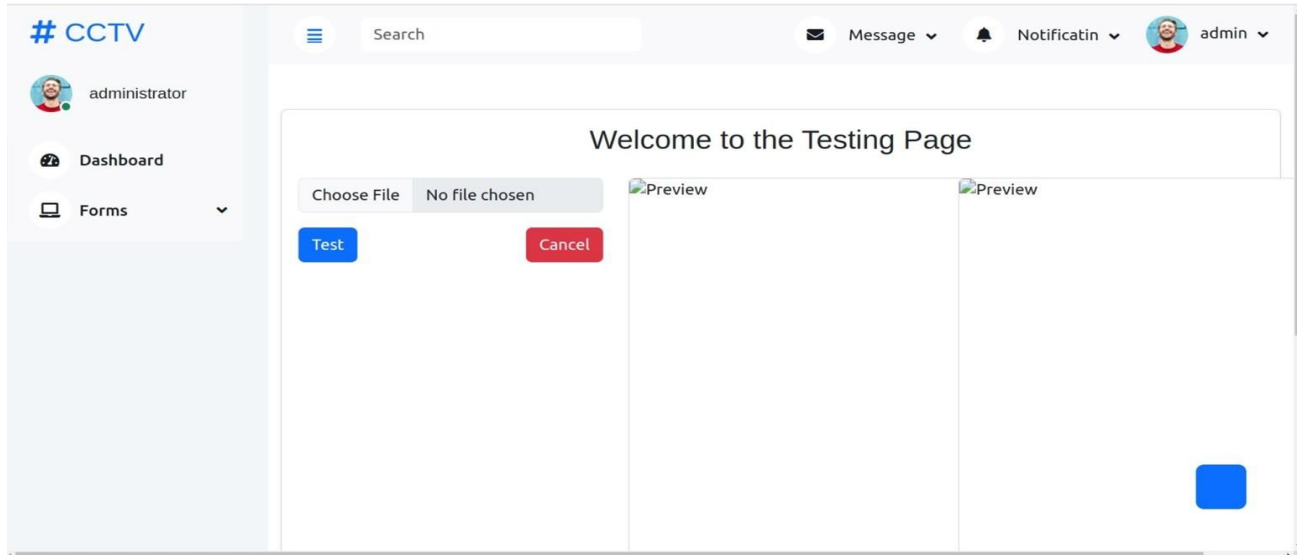


Figure 18:5:4: Testing page 1

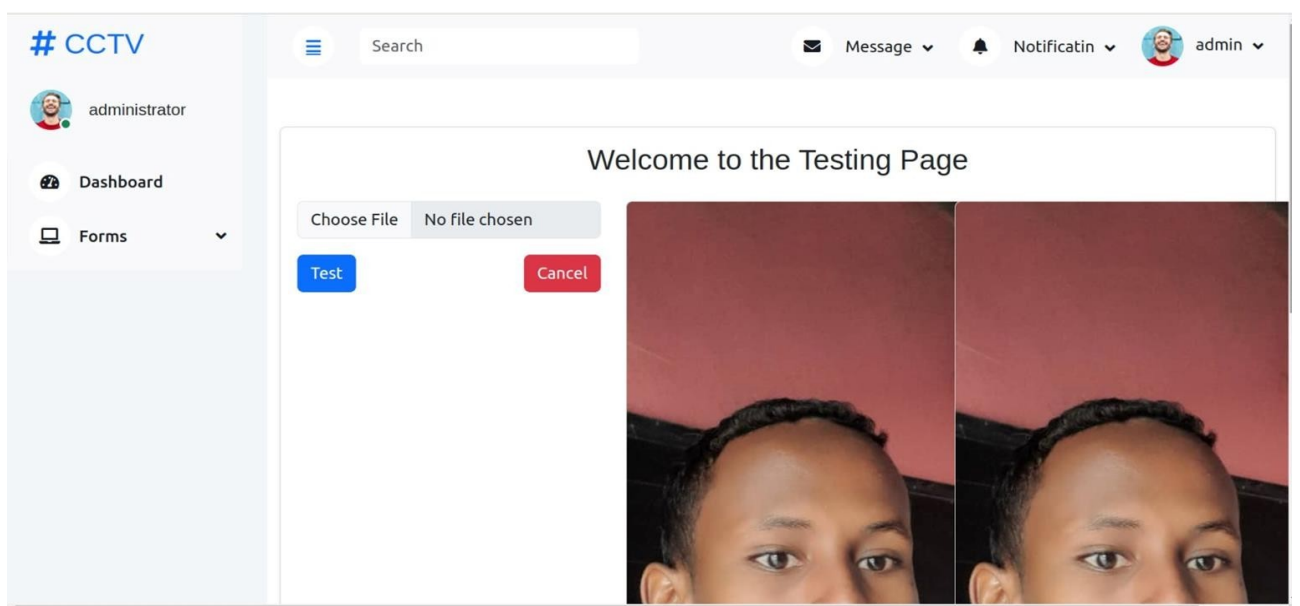


Figure 19:5:5: testing page 1

- The testing page allows users to upload files containing images for testing purposes. Users can select files from their local storage and submit them for processing.
- In this snapshot, users can upload existing image files for testing. The system performs facial recognition and classification in real-time, displaying the results promptly.

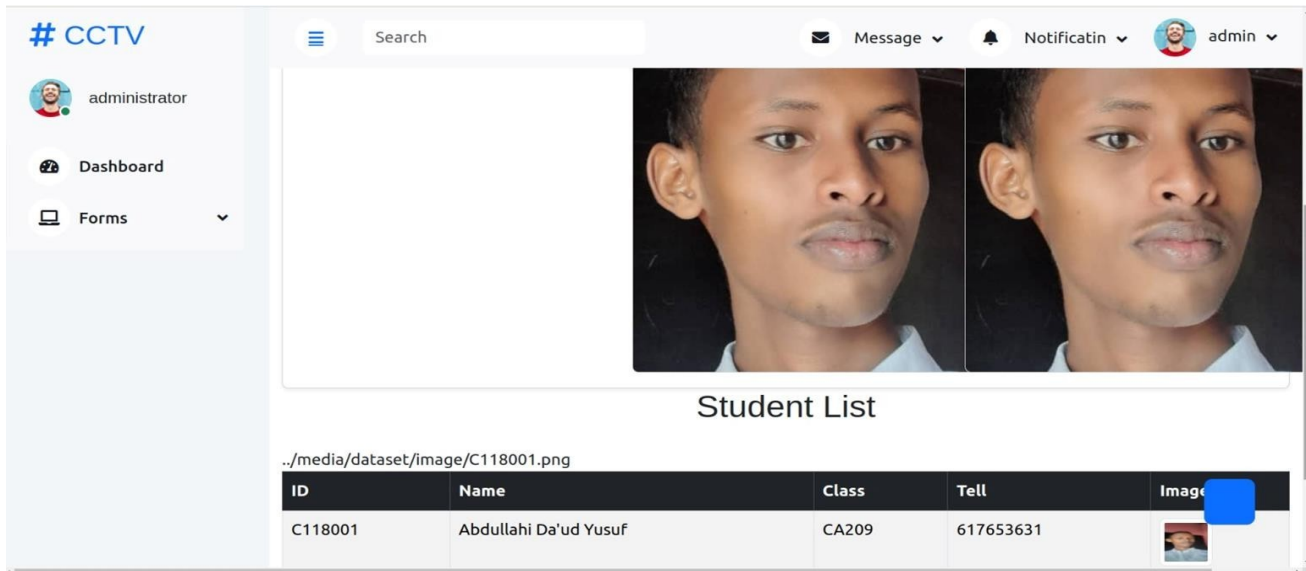


Figure 20:5:6: Result page

- The student list page provides an overview of all registered students, including their names, IDs, classes, and associated images. Users can search, filter, or sort the list to find specific students quickly.

These snapshots offer a glimpse into the functionality and user interface of the implemented system, demonstrating its effectiveness in identifying and classifying students based on facial recognition technology. The next chapter will delve into the evaluation and validation of the system to assess its accuracy, reliability, and usability in real-world scenarios.

CHAPTER 6: DISCUSSION AND RESULTS

6.0 Discussions

The development of an image-based student identification and classification system using face recognition technology represents a significant innovation in university building security. By leveraging advanced facial recognition algorithms, this system provides an efficient and accurate method for identifying and classifying students based on images captured by CCTV cameras installed within educational institutions.

A primary advantage of this system is its ability to process images captured from CCTV footage to differentiate and lawful students. Unlike traditional methods, which may involve manual checking was to ask the teachers or staff of the university which class is this Image if they recognize, this system can analyze uploaded captured CCTV images to determine the which was or the suspect of students it also saves more time for investigator.

Moreover, the system addresses the challenge of identifying individuals captured by CCTV cameras. By processing images of single individuals from different angles and under diverse lighting conditions, the system can effectively recognize faces, making it highly useful for monitoring student populations during various campus activities.

Privacy is one of main goal the information of the Like student ID, student name, student class, student phone and so on stores inside the Image while Image encoding and ID also stores pickle file to keep responsibility of student information.

6.1 Results

- Based on the discussion provided, here are the key results:

Innovation in University Security: The development of an image-based student identification and classification system using face recognition technology represents a significant innovation in university building security. This system leverages advanced facial recognition algorithms to

efficiently and accurately identify and classify students based on CCTV footage captured within educational institutions.

- Advantages of the System:

Efficiency: Automates the identification process, distinguishing between and lawful students without manual intervention.

Accuracy: Uses advanced algorithms to process CCTV images and recognize faces, overcoming challenges posed by different angles and lighting conditions.

Time-saving: Saves time for investigators compared to traditional manual methods or faculty recognition.

- Challenges and Limitations:

Resource Constraints: Limited time and resources during the research phase may have impacted the system's thoroughness and development.

Technological Imperfections: Recognizes the inherent imperfections in any technological solution, requiring ongoing adjustments and refinements for optimal performance.

- Future Directions:

Research and Development: Emphasizes the need for continued research to address limitations and enhance the system's effectiveness and scalability.

Security Enhancement: Highlights the potential for further innovations in enhancing campus security through technological advancements.

In summary, while the system shows promising advancements in university security through automated face recognition technology, ongoing research and development are crucial to optimize its capabilities and address inherent limitations.

6.2 Accomplishment of the objectives

We completed the project by designing and building a Image-based Student Identification and Classification using Face Recognition system that allows to Recognize student into lawful students.

CHAPTER 7: Conclusion and Future Work

7.0 Introduction

In this chapter we clarify the conclusion of the exploration that has been conducted by the research in nine months, this chapter discusses key points such as the conclusion of the research, the achievement of the research objectives that we have already mentioned in chapter one of the research, recommendations, and future work for those who wish to conduct similar research.

7.1 Limitation of the Study

Although the study was completed efficaciously by considering all the possible solutions human work is always never perfect and has some weaknesses. This research has a number of challenges mainly caused by limited time and other challenges faced by the researchers during the study. In other words, if you think that your system is complete, it is not perfect as others will think that your system needs some configurations and changes.

7.2 Recommendation and Future work

As we stated above in this chapter, we by our best to make the system successfully but it is possible to remember something that has more importance to the system to enhance its functionality, everyone who wants to make enhancement on the system (editing and modifying code of the system) is allowable to proceed to keep the privacy and copyright in this section we also state some of the important element that left as future work and recommendation because of the limit time we faced. These points we did not able to accomplish in the system include:

- To make accuracy about 95% or above to become more reliable.
- To recognize by eye instead of face that is more important.

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APPENDIX MODEL

```
import os,random,face_recognition,pickle,numpy as np,cv2

from pathlib import Path

import Face_Recognition.xog as xog

import os

from PIL import Image

from stegano import lsb

pathka = os.path.dirname(os.path.dirname(os.path.abspath(_file_)))

db=xog.Data()

db.create_table()

class Main:

def __init__(self,

ddir=pathka + "/media/dataset/image/",

data=pathka + "/media/dataset/data/",

test=pathka + "/media/dataset/test/",

folder_path=pathka + "/media/dataset/check/"):

self.image_extensions = ['.jpg', '.jpeg', '.png', '.gif'] # Add more extensions if needed

self.ddir=ddir

self.folder_path=folder_path

self.data=data

self.test=test

if not os.path.exists(self.folder_path):

try:

os.makedirs(self.ddir)

os.makedirs(self.data)

os.makedirs(self.test)
```

```

except:

pass

def get_num_of_student(self):

num =len([file for file in os.listdir(self.ddir)])

return num

def hide_data_on_image(self,image,data):

secret = lsb.hide(str(image),data)

image = image[:-3]+"png"

secret.save(image)

delete = image[:-3]+"jpg"

os.remove(delete)

def show_data_on_image(self,image):

clear_message = lsb.reveal(image)

clear_message = clear_message.split(",")

imageka =clear_message[-1]

imageka = imageka.split("/Group76/")[1]

imageka = "../"+imageka[:-3]+"png"

clear_message[-1]=imageka

return clear_message

def crop_face_and_neck(self,input_path, output_path):

image = face_recognition.load_image_file(input_path)

face_locations = face_recognition.face_locations(image)

if face_locations:

top, right, bottom, left = face_locations[0]

height = bottom - top

new_bottom = int(bottom + 0.5 * height)

```

```

pil_image = Image.open(input_path)

face_image = pil_image.crop((left, top, right, new_bottom))

face_image.save(output_path)

else:

return "No faces found in the image."

def count_files_in_folder(self,folder_path):

file_count = 0

for _, _, files in os.walk(folder_path):

file_count += len(files)

return file_count

def rename_files(self,folder_path=pathka + "/media/dataset/check/"):

self.folder_path=folder_path

self.image_files = []

for file_name in os.listdir(self.folder_path):

if os.path.isfile(os.path.join(self.folder_path, file_name)):

_, extension = os.path.splitext(file_name)

if extension.lower() in self.image_extensions:

self.image_files.append(file_name)

return len(self.image_files)

def generate(self,student_info):

first_male = ["Liban", "khadar", "Abas", "suleyman", "Omar", "Abdullahi"]

first_female = ["khadro", "sacdiyo", "xalimo", "caasho", "safiyo", "ruweydo"]

second_names = ["Yaxye", "mohamed", "Axmed", "Maxmuud", "Abdiqaadir",

"Da'ud"] last_names = ["Cali", "Geedi", "Yusuf", "Kaafi", "Abdulle", "Khaalid"]

fasal = ["CA206", "CA207", "CA209", "CA201", "CA202", "CA205"]

try:

```

```

pickle_file_path = self.data+"encodings.pkl"

with open(pickle_file_path, 'rb') as file:

data = pickle.load(file)

encode = data["encodings"]

ID = data ["ID"]

except:

encode = []

ID = []


last_first_male_index = len(first_male) - 1

last_second_names_index = len(second_names) - 1

last_last_names_index = len(last_names) - 1

DEFAULT_ENCODINGS_PATH = Path(self.data+"encodings.pkl")

count_test=0

count_remove=0

count=0

saved=0

kii=self.get_num_of_student()

for file_name in os.listdir(self.folder_path):

if os.path.isfile(os.path.join(self.folder_path, file_name)):

_, extension = os.path.splitext(file_name)

if extension.lower() in self.image_extensions:

first = random.randint(0, last_first_male_index)

second = random.randint(0, last_second_names_index)

last = random.randint(0, last_last_names_index)

full_name = first_male[first] + " " + second_names[second] + " " + last_names[last]

```

```

fasalka = fasal[last]

file = self.folder_path + file_name

try:

image = face_recognition.load_image_file(file)

face_encoding = face_recognition.face_encodings(image)

encode_test=np.array(encode)

except:face_encoding=[]

if len(face_encoding) > 0

:

face_distances = face_recognition.face_distance(encode_test, face_encoding[0])

matches = face_distances < 0.2

face_distances_test = face_recognition.face_distance(encode_test, face_encoding[0])

matches_test = face_distances_test < 0.4

if np.any(matches):

count+=1

elif np.any(matches_test):

count_test+=1

else:

kii+=1

if len(student_info) != 0:

IDga ,full_name,fasal,phone = student_info[0] ,student_info[1],student_info[2],student_info[3]

else:

IDga = "C11800" + str(kii)

phone = "617653631"

ID.append(IDga)

encode.append(face_encoding[0])

```

```
new_file = self.ddir+IDga+".jpg"
```

```

sawir = new_file.replace('/home/liban/Downloads/Group7676', '')
os.rename(self.folder_path+file_name,new_file)
self.crop_face_and_neck(new_file,self.test+IDga+".jpg")
xogta = IDga+","+full_name+","+fasalka+","+str(phone)+","+sawir
self.hide_data_on_image(new_file,xogta)

saved+=1

else:

count_remove+=1

try:

os.remove(self.folder_path+file_name)

except:

pass

name_encodings = {"ID": ID, "encodings": encode}

with DEFAULT_ENCODINGS_PATH.open(mode="wb") as f:

pickle.dump(name_encodings, f)

return "saved images : "+str(saved), "we catched duplicated images are :"+str(count)+" ", "Same
person images are "+str(count_test), "remove images that was not found face are :

"+str(count_remove)

def load_data(self):

# Read pickle data

try:

pickle_file_path = self.data+"encodings.pkl"

with open(pickle_file_path, 'rb') as file:

data = pickle.load(file)

self.known_face_encodings =

data["encodings"] self.known_face_names =

```

data["ID"]

```

self.image_list = db.get_column("Image ")

except:

self.known_face_encodings = []

self.known_face_names = []

self.image_list = db.get_column("Image

")

def recognize_face(self, test_image_path, **kwargs):

self.load_data()

distance_threshold=kwargs["start"]

closed = kwargs["closed"]

fileka=test_image_path.split("/")[-1:]

fileka = self.data+fileka[0]

x = self.crop_face_and_neck(test_image_path,fileka)

if x != "No faces found in the image.":

test_image = face_recognition.load_image_file(fileka)

test_face_encodings = face_recognition.face_encodings(test_image)

else:

return "No face found in the test image"


face_distances = face_recognition.face_distance(self.known_face_encodings,

test_face_encodings[0])

matches = face_distances < distance_threshold

if np.any(matches):

matched_names = np.array(self.known_face_names)[matches].tolist()

all = []

for i in matched_names:

```

```
i= self.ddir+i+".png"
```

```
x = self.show_data_on_image(i)

all.append(x)

return all

elif distance_threshold<closed:

start=distance_threshold+0.1

x = self.recognize_face(test_image_path,start=start,closed=closed)

return x

else:

os.remove(fileka)

return ("No match found for the test image.")
```

APPENDIX B CONTROL

```
# student_module.py

import sqlite3

from sqlite3 import Error

class Data:

    def __init__(self):

        """Initialize the connection to the SQLite database."""

        self.db_file = "just.db"

        self.conn = self.create_connection()

        self.create_user()

    def create_connection(self):

        """Create a database connection to the SQLite database."""

        conn = None

        try:

            conn = sqlite3.connect(self.db_file, check_same_thread=False) # Allow thread sharing

        except Error as e:

            print(e)

        return conn

    def count(self):

        count="select count(*) from Student "

        try:

            c = self.conn.cursor()

            cou=c.execute(count)

            count=c.fetchone()[0]

        return count

    except:
```

```

pass

def create_table(self):
    """Create the Student table if it does not exist."""
    create_table_sql = """
    CREATE TABLE IF NOT EXISTS Student
    ( ID text PRIMARY KEY,
    Name text NOT NULL,
    Class text NOT NULL,
    Telephone text NOT NULL,
    Image text NOT NULL
    );
    """

    try:
        c = self.conn.cursor()
        c.execute(create_table_sql)

    except Error as e:
        print(e)

    def get_number(self,table):
        count="select count(*) from "+table

        try:
            c = self.conn.cursor()
            cou=c.execute(count)
            count=c.fetchone()[0]

        return count

    except:

pass

```

```

def delete_user(self,id):

sql = "DELETE FROM Users WHERE User = ?"

try:

cur = self.conn.cursor()

cur.execute(sql, (id,))

self.conn.commit()

except Error as e:

print(e)

def create_user(self):

"""Create the User table if it does not exist."""

create_table_sql = """

CREATE TABLE IF NOT EXISTS Users (

User text NOT NULL unique,

Gmail text NOT NULL unique,

user_type not null,

Pass text NOT NULL

);

"""

try:

c = self.conn.cursor()

c.execute(create_table_sql)

c.execute("insert      into      Users      (User,Gmail,user_type,Pass)      values

('administrator','administrator@gmail.com','admin','1234')")

except Error as e:

print(e)

return e

```

```

def insert_data(self, ID, name, student_class, telephone, image):
    """Insert a new row into the Student table."""
    sql = "INSERT INTO Student (ID, Name, Class, Telephone, Image) VALUES (?, ?, ?, ?, ?)"
    try:
        cur = self.conn.cursor()
        cur.execute(sql, (ID, name, student_class, telephone, image))
        self.conn.commit()
    except Error as e:
        print(e)

def insert_user(self, User, gmail,user_type, Pass):
    """Insert a new row into the Student table."""
    sql = "INSERT INTO Users (User, gmail, user_type,Pass) VALUES (?, ?, ?, ?)"
    try:
        cur = self.conn.cursor()
        cur.execute(sql, (User, gmail,user_type, Pass))
        self.conn.commit()
    except Error as e:
        return e

def update_data(self, ID, name=None, student_class=None, telephone=None, image=None):
    """Update rows in the Student table."""
    updates = []
    params = []

    if name:
        updates.append("Name = ?")
        params.append(name)

```

```

if student_class:
    updates.append("Class = ?")
    params.append(student_class)
    if telephone:
        updates.append("Telephone =
        ?") params.append(telephone)
    if image:
        updates.append("Image = ?")
        params.append(image)

    if updates:
        updates_str = ', '.join(updates)
        params.append(ID)

    sql = f"UPDATE Student SET {updates_str} WHERE ID = ?"
    try:
        cur = self.conn.cursor()
        cur.execute(sql, tuple(params))
        self.conn.commit()
    except Error as e:
        print(e)
    else:
        print("No updates provided.")

def delete_data(self, student_id):
    """Delete rows from the Student table."""

```

```

sql = "DELETE FROM Student WHERE ID = ?"

try:

cur = self.conn.cursor()

cur.execute(sql, (student_id,))

self.conn.commit()

except Error as e:

print(e)

def get_user(self,user=None):

conditions = []

params = []

if user:

conditions.append("User = ?")

params.append(user)

conditions_str = ' AND '.join(conditions) if conditions else '1=1'


sql = f"SELECT * FROM Users WHERE {conditions_str}"

try:

cur = self.conn.cursor()

cur.execute(sql, tuple(params))

rows = cur.fetchall()

List = []

for i in rows:

for n in i:

List.append(n)

return List

except Error as e:

```

```
return e
```

```
def search_data(self, ID=None, name=None, student_class=None, telephone=None, Image=None):
```

```
    """Search for rows in the Student table."""
```

```
    conditions = []
```

```
    params = []
```

```
    if ID:
```

```
        conditions.append("ID = ?")
```

```
        params.append(ID)
```

```
    if name:
```

```
        conditions.append("Name = ?")
```

```
        params.append(name)
```

```
    if student_class:
```

```
        conditions.append("Class =
```

```
        ?")
```

```
        params.append(student_class)
```

```
    if telephone:
```

```
        conditions.append("Telephone =
```

```
        ?") params.append(telephone)
```

```
    if Image:
```

```
        conditions.append("Image = ?")
```

```
        params.append(Image)
```

```
    conditions_str = ' AND '.join(conditions) if conditions else '1=1'
```

```
    sql = f"SELECT * FROM Student WHERE {conditions_str}"
```

```
    try:
```

```
        cur = self.conn.cursor()
```

```
cur.execute(sql, tuple(params))
```

```

rows = cur.fetchall()

List = []

for i in rows:

    for n in i:

        if "/media/liban/9b8fb9f1-4b35-4398-b159-67acf64bc182/liban/Fiid/Group76" in n:

            n = n.replace("/media/liban/9b8fb9f1-4b35-4398-b159-67acf64bc182/liban/Fiid/Group76","../")

            List.append(n)

    return List

except Error as e:

    return []

def get_column(self,col):

    sql = "Select " + col + "from Student"

    try:

        cur = self.conn.cursor()

        cur.execute(sql)

        rows = cur.fetchall()

        modify_row = [item[0] for item in rows]

        return modify_row

    except Error as e:

        print(e)

        return []

def get_users(self):

    try:

        cur = self.conn.cursor()

        cur.execute("select * from Users")

        rows = cur.fetchall()

```

```
List = []

for i in rows:

    List.append(i)

return List

except Error as e:

    print(e)

return []

def get_students(self):

    try:

        cur = self.conn.cursor()

        cur.execute("select * from Student ")

        rows = cur.fetchall()

        List = []

        for i in rows:

            List.append(i)

        return List

    except Error as e:

        print(e)

    return []
```

APPENDIX C DATA BASE

```
from django.shortcuts import render,redirect

from django.http import request,HttpResponse

from django.http import JsonResponse

from django.contrib.auth import logout

from django.utils.decorators import decorator_from_middleware

from Group76.middleware import NoCacheMiddleware

import os

pathka = os.path.dirname(os.path.dirname(os.path.abspath(_file_)))

from .main import *

from .xog import *

from .models import *

db.create_user()

obj=Main()

def login(request):

return render(request,"login.html",{"user":""})

def check(request):

if request.method ==

"POST": user =

request.POST ["user"] Pass =

request.POST ["pass"] userka

= db.get_user(user) try:

if user == userka[0] and Pass == userka[3]:

request.session["user"] = userka[0]

request.session["type"] = userka[2]
```

```

return render(request,"index.html",

{"userka":db.get_number("Users"),"students":obj.get_num_of_student(

)})

else:

userka = "User or Password is incorrect"

except:

userka = "This User user is not exists"

return render(request,"login.html",{"user":userka})

def edit(request, id):

print(request.method == "POST")

if request.method == "POST":

username = request.POST["user"]

password = request.POST["pass"]

email = request.POST["gmail"]

user_type = request.POST["type"]

if email.endswith("@gmail.com"):

db.delete_user(id)

result = db.insert_user(username, email, user_type, password)

print(result is None)

if result is None:

return render(request, "signup.html", {"fill": [""], "users": db.get_users()})

else:

return render(request, "signup.html", {"error": result, "users": db.get_users()})

else:

return render(request, "signup.html", {"error": "invalid gmail", "users": db.get_users()})

else:

```

```

user_details = db.get_user(id)

users = db.get_users()

return render(request, "signup.html", {"fill": user_details, "users": users})

def save(request):

if request.method == "POST":

user = request.POST ["user"]

Pass = request.POST ["pass"]

gmai = request.POST ["gmail"]

user_type = request.POST ["type"]

if gmai[-10:] == "@gmail.com":

x = db.insert_user(user,gmai,user_type,Pass)

if x == None:

return render(request,"signup.html",{"fill":[""], "users":db.get_users()})

else:

return render(request,"signup.html",{"error":x , "users":db.get_users()})

else:

return render(request,"signup.html",{"error":"invalid gmail", "users":db.get_users()})

def index(request):

return render(request,"index.html",

{"students":obj.get_num_of_student(),"userka":db.get_number("Users"

)})

def train(request):

return render(request,"train.html",{"num_of_image":{"header":"","content":"","info_of_image":

{"dup_h

eader":"","dup_content":"","remove_header":"","rem_content":""}})

def test(request):

```

```

return render(request,"test.html",{ })

def signup(request):

return render(request,"signup.html",{ "fill":[""],"users":db.get_users()})

def logout(request):

response = redirect('login')

response['Cache-Control'] = 'no-store, no-cache, must-revalidate, max-age=0'

response['Pragma'] = 'no-cache'

response['Expires'] = '0'

return response

def rename(request):

if request.method=="POST":

try:

for i in os.listdir(pathka + "/media/dataset/check/"):

os.remove(pathka +"/media/dataset/check/"+i)

except:

pass

images=request.FILES.getlist("Images")

try:

id = request.POST["ID"]

Name = request.POST["Name"]

Class = request.POST["Class"]

Email = request.POST["Email"]

except:

id=""

img=Images()

for i in images:

```

```

img.images=i

img.save()

try:

num_of_files=obj.rename_files()

num_of_files="number of Images of renamed are : "+str(num_of_files)

if id != "":

info_of_image=obj.generate([id,Name,Class,Email])

else:

info_of_image=obj.generate([])

except:pass

return render(request,"train.html",{"num_of_image":{"header":"Saved
Image","content":info_of_image[0]},"info_of_image":{"dup_header":"duplicated images and same
person",
"dup_content":info_of_image[2]+"\\t\\n"+info_of_image[1],"remove_header":"removed
Image","rem_content":info_of_image[3]}}))

def recognize_faces(request):

List =[]

sawir = ""

if request.method=="POST":

try:

for i in os.listdir(pathka + "/media/dataset/check/"):

os.remove(pathka + "/media/dataset/check/"+i)

except:

pass

images=request.FILES.getlist("Images")

```

```

img=Images()

for i in images:

img.images=i

img.save()

for i in os.listdir(pathka + "/media/dataset/check/"):

sawir = i

x=obj.recognize_face(pathka + "/media/dataset/check/"+i,start=0.1,closed=0.35)

if x == "No match found for the test image.":

x2=obj.recognize_face(pathka + "/media/dataset/check/"+i,start=0.35,closed=0.5)

List.append(x2)

else:

List.append(x)

print(List)

return render(request,"test.html",{ "test_image":"/media/dataset/check/"+sawir,"data":List[0]})

else:

return render(request,"test.html",{ })

```